

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Analytical Problem Solving**

**COURSE NAME:** Cloud Computing and Big  
Data Analytics

**COURSE CODE:** CSA1507

---

**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation(Level2)  
Question Count: 5

**Total Marks: 20**

---

**Code of Conduct**

I, S.Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Susannal Devapriyam.

**Analytical Problem Solving Rubric**

| Criteria                 | Weightage | Level 4 - Exemplary                                             | Level 3 – Proficient                   | Level 2 - Developing            | Level 1 - Beginning         |
|--------------------------|-----------|-----------------------------------------------------------------|----------------------------------------|---------------------------------|-----------------------------|
| Problem Interpretation   | 20%       | Interprets all technical requirements accurately                | Interprets most requirements correctly | Partial interpretation          | Misinterprets the problem   |
| Analytical Reasoning     | 25%       | Strong reasoning with clear cause-effect analysis               | Good reasoning with minor gaps         | Basic analysis only             | Weaken analysis             |
| Method Selection         | 20%       | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts       | Partially suitable concepts     | Inappropriate approach      |
| Solution Quality         | 20%       | Provides feasible and technically sound solution                | Reasonable solution with minor issues  | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15%       | Well-structured and technically precise                         | Generally clear                        | Somewhat unclear                | Difficult to follow         |

**Total Marks Awarded :**

## **Q1. Design an Identity Verification Process for a Cloud-Based Online Examination System**

### **Introduction**

With the rapid growth of cloud computing, educational institutions are increasingly conducting examinations through online platforms. Cloud-based online examination systems allow students to attend exams from any location using a computer or mobile device connected to the Internet. Although these systems provide flexibility and convenience, they also introduce security challenges. One of the biggest challenges is ensuring that the person taking the examination is the actual registered student.

Identity verification is a process that confirms the identity of a user before allowing access to the examination. A well-designed identity verification system prevents impersonation, cheating, and unauthorized access. Modern cloud-based examination systems use multiple authentication methods such as passwords, One-Time Passwords (OTP), biometric verification, facial recognition, and artificial intelligence (AI)-based monitoring to improve security. These methods work together to ensure that only genuine students are allowed to participate in online examinations.

### **What is Identity Verification?**

Identity verification is the process of confirming that a person is truly who they claim to be. In an online examination system, it ensures that the registered student is the one writing the examination.

Unlike traditional classroom examinations, where an invigilator physically checks the student's identity card, online examinations require digital methods to verify identity. These methods use secure authentication techniques before and during the examination.

The main purpose of identity verification is to maintain the fairness, integrity, and security of online examinations.

### **Need for Identity Verification in Cloud-Based Online Examinations**

Identity verification is an important part of every online examination system because it helps maintain trust and fairness. Without proper verification, anyone could access another student's account and write the examination on their behalf.

The major reasons for identity verification include:

- Preventing impersonation and unauthorized access.
- Ensuring that only registered students can take the examination.
- Protecting confidential examination questions.
- Maintaining academic integrity.
- Reducing cheating during online examinations.
- Protecting student records stored in the cloud.

A secure verification system increases confidence among students, teachers, and educational institutions.

### **Identity Verification Process**

A cloud-based online examination system should follow several verification steps before allowing the student to begin the examination.

#### **Step 1: Student Registration**

The first step is student registration.

Each student creates an account by providing personal information such as:

- Full Name
- Student Registration Number
- College Email ID
- Mobile Number
- Date of Birth

The student also uploads:

- Passport-size photograph
- College Identity Card
- Signature

All this information is securely stored in the cloud database.

#### **Step 2: Login Authentication**

Before the examination begins, the student logs into the examination portal using a username and password.

The system verifies the entered credentials with the stored records.

Passwords are encrypted so that they cannot be easily stolen or viewed by unauthorized users.

If the username or password is incorrect, access to the examination is denied.

### **Step 3: Multi-Factor Authentication (OTP Verification)**

After entering the correct password, the system performs an additional security check using Multi-Factor Authentication (MFA).

A One-Time Password (OTP) is sent to the student's registered mobile number or email address.

The student must enter the correct OTP within a limited time.

Only after successful OTP verification is the student allowed to proceed.

This additional step prevents attackers from accessing the examination even if they know the student's password.

### **Step 4: Face Verification**

The examination system requests permission to access the student's webcam.

A live photograph is captured and compared with the photograph uploaded during registration.

Artificial Intelligence (AI) and facial recognition algorithms analyze important facial features such as:

- Eyes
- Nose
- Mouth
- Face shape

If both images match, identity verification is successful.

If the face does not match, the system may request another verification attempt or deny access.

### **Step 5: Student ID Card Verification**

The student is asked to display the college identity card in front of the webcam.

The system checks important details such as:

- Student Name
- Registration Number
- Photograph

Some systems also use Optical Character Recognition (OCR) to automatically read the information printed on the ID card.

This step provides an additional layer of authentication.

### **Step 6: Exam Environment Verification**

Before the examination starts, the system checks the student's surroundings.

The webcam scans the room to ensure:

- Only one person is present.
- No books or mobile phones are visible.
- No unauthorized materials are available.

The microphone may also monitor background sounds.

This helps create a secure examination environment.

### **Step 7: Continuous AI-Based Monitoring**

Identity verification should not stop after login.

During the examination, Artificial Intelligence continuously monitors the student's activities.

The system checks for suspicious behavior such as:

- Looking away from the screen repeatedly.

- Another person entering the room.
- Multiple faces detected.
- Frequently switching browser tabs.
- Using unauthorized applications.

If suspicious activity is detected, the system immediately alerts the examiner.

### **Step 8: Secure Cloud Storage**

All examination activities are securely stored in the cloud.

The stored information includes:

- Login time
- OTP verification
- Face verification results
- Examination logs
- Recorded video
- Exam answers

Sensitive data is protected using encryption to prevent unauthorized access. Advantages

A well-designed identity verification process offers many advantages.

- Prevents impersonation during examinations.
- Ensures that only authorized students can access the exam.
- Protects confidential examination data.
- Reduces cheating through continuous monitoring.
- Improves trust in online examination systems.
- Maintains fairness and academic integrity.
- Provides secure storage of student information.

### **Challenges**

Despite its benefits, identity verification also has some challenges.

- Poor internet connectivity may interrupt verification.

- Face recognition may fail due to poor lighting or camera quality.
- Students may have privacy concerns regarding webcam monitoring.
- Implementing AI and biometric systems increases cost.
- Technical failures may delay examination access.

### **Real-World Example**

Many universities and certification organizations use cloud-based online examination systems with identity verification.

For example, certification platforms such as **Pearson VUE** and **PSI Online** verify candidates using government ID cards, live facial verification, webcam monitoring, and AI-based proctoring before allowing them to start the examination. These methods ensure that the registered candidate is taking the exam honestly.

### **Conclusion**

Identity verification is one of the most important security features in a cloud-based online examination system. By combining username and password authentication, OTP verification, facial recognition, ID card verification, AI monitoring, and secure cloud storage, educational institutions can conduct online examinations safely and fairly. A strong identity verification process protects student identities, prevents cheating, and ensures the credibility of online examinations.

## Q2. Evaluate How Digital Certificates Improve Authentication

### Introduction

In today's digital world, people use the Internet for banking, online shopping, education, cloud computing, and many other services. Since sensitive information such as passwords, personal details, and financial data is shared online, it is important to ensure that users are communicating with trusted websites and servers. This is where **digital certificates** play an important role.

A digital certificate helps verify the identity of a website, user, or organization. It creates trust between users and cloud services by ensuring that communication takes place with a genuine and authorized party. Digital certificates also support secure authentication and protect data from cyber-attacks.

### What is a Digital Certificate?

A **digital certificate** is an electronic document that proves the identity of a website, user, server, or organization on the Internet. It is similar to an identity card or passport in the digital world.

A digital certificate contains information such as the owner's name, public key, certificate validity period, issuing authority, and digital signature. It is issued by a trusted organization called a **Certificate Authority (CA)**.

Whenever a user connects to a secure website, the digital certificate helps verify that the website is genuine and that the communication is secure.

### Need for Digital Certificates

Digital certificates are necessary because the Internet is an open network where attackers may try to steal information or create fake websites.

Without digital certificates:

- Users cannot easily verify whether a website is genuine.
- Passwords and personal information can be stolen.
- Hackers can perform phishing attacks.
- Communication may not be secure.



Digital certificates solve these problems by verifying identities and protecting online communication.

## **Components of a Digital Certificate**

A digital certificate contains several important details.

### **1. Certificate Owner**

The name of the website, organization, or user to whom the certificate belongs.

### **2. Public Key**

A public encryption key used for secure communication.

### **3. Certificate Authority (CA)**

The trusted organization that issues the certificate.

### **4. Validity Period**

Shows the issue date and expiry date of the certificate.

### **5. Digital Signature**

The electronic signature of the Certificate Authority that proves the certificate is genuine.

## **How Digital Certificates Work**

The authentication process using digital certificates follows several steps.

### **Step 1: User Visits a Website**

A user opens a secure website such as an online banking website or cloud application.

### **Step 2: Website Sends its Digital Certificate**

The website automatically sends its digital certificate to the user's browser.

### **Step 3: Certificate Verification**

The browser checks:

- Whether the certificate is issued by a trusted Certificate Authority.
- Whether the certificate has expired.
- Whether the certificate belongs to the correct website.

If everything is valid, the browser trusts the website.

#### **Step 4: Secure Connection Established**

The browser and server create an encrypted communication channel using SSL/TLS protocols.

Now all information exchanged between them is protected.

#### **Step 5: Secure Authentication**

The user safely enters usernames, passwords, or payment information without worrying that someone will steal the data during transmission.

#### **Role of Certificate Authority (CA)**

A **Certificate Authority (CA)** is a trusted organization responsible for issuing and managing digital certificates.

The CA verifies the identity of the website or organization before issuing a certificate.

Some popular Certificate Authorities include:

- DigiCert
- GlobalSign
- Sectigo
- Let's Encrypt

Because browsers trust these Certificate Authorities, users can also trust the websites that receive certificates from them.

#### **How Digital Certificates Improve Authentication**

Digital certificates improve authentication in many ways.

### **1. Verify Identity**

Digital certificates confirm that the website or user is genuine.

This prevents users from connecting to fake websites

### **2. Enable Secure Login**

Many cloud services use digital certificates during login authentication.

Only trusted servers can communicate with authenticated users.

### **3. Protect Data Using Encryption**

Digital certificates help establish encrypted communication.

Even if hackers intercept the data, they cannot read it because it is encrypted.

### **4. Prevent Phishing Attacks**

Cybercriminals often create fake websites to steal passwords.

Digital certificates help users identify genuine websites by displaying HTTPS and the security lock symbol.

### **5. Ensure Data Integrity**

Digital certificates make sure that data is not modified while travelling across the Internet.

If any information changes during transmission, the system immediately detects it.

### **6. Build Trust**

When users see a secure HTTPS connection and a valid certificate, they feel confident that the website is safe.

This increases trust between customers and organizations.

## **Digital Certificates in Cloud Computing**

Cloud service providers use digital certificates for many purposes.

Some common uses include:

- Secure user login
- Cloud server authentication
- API authentication
- Secure email communication
- VPN authentication
- Protecting cloud databases
- Secure communication between cloud applications

Without digital certificates, cloud security would become much weaker.

### **Real-World Applications**

Digital certificates are widely used in everyday life.

Some examples include:

- Online Banking
- Amazon
- Gmail
- Microsoft Azure
- Google Cloud Platform
- AWS Management Console
- Government websites
- University online examination portals

Whenever you see **https://** and a **lock icon** in the browser, the website is using a digital certificate.

### **Advantages of Digital Certificates**

Digital certificates provide many benefits.

- Verify the identity of websites and users.
- Protect sensitive information.

- Encrypt communication.
- Prevent phishing attacks.
- Ensure secure authentication.
- Increase customer trust.
- Support secure cloud computing.
- Maintain data integrity.

### **Limitations of Digital Certificates**

Although digital certificates improve security, they also have some limitations.

- Certificates expire and must be renewed.
- Managing certificates can be expensive.
- If a private key is stolen, attackers may misuse the certificate.
- Organizations must properly manage and update certificates.
- Revoked certificates must be replaced immediately.

### **Conclusion**

Digital certificates are an essential part of modern cloud security. They verify the identity of websites, users, and servers while protecting data through encryption. By preventing phishing attacks, ensuring secure authentication, and building trust between users and cloud services, digital certificates make online communication safe and reliable. As cloud computing continues to grow, the importance of digital certificates in protecting digital identities and maintaining secure communication will continue to increase.